

PROBLEM-SOLUTION FIT - TEAM 154

Project: Insurance fraud detection using Machine Learning
Max Marks: 4 Marks

DEFINE CS, FIT INTO CC

1. CUSTOMER SEGMENT(S)

CS

Who is your customer? i.e. working parents of 0-5 y.o. kids

Primary customers:

- Insurance companies
- Claims verification departments
- Investigators & risk teams

Secondary customers:

- Financial institutions
- Government regulatory authorities
- Insurance data analysts

Needs automated system for large numbers of claims.

6. CUSTOMER CONSTRAINTS

CC

What constraints prevent your customers from taking action or limit their choices of solutions?

- Limited technical AI knowledge
- Budget constraints & integration challenges
- Data privacy and security concerns
- Resistance to adopting new technology

Requirement: Must be easy to integrate and cost-effective.

5. AVAILABLE SOLUTIONS

AS

Which solutions are available to the customers when they face the problem or need to get the job done?

Current Options:

- Manual verification
- Rule-based systems
- Statistical analysis
- Internal teams

Key Problems:

- Time-consuming & high operational cost
- Limited capability
- Human bias/error

Advantage: ML provides more accurate, automated solution.

EXPLORE AS, DIFFERENTIATE

FOCUS ON J&P TAP INTO BE, UNDERSTAND RC

2. JOBS-TO-BE-DONE / PROBLEMS

J&P

Which jobs-to-be-done (or problems) do you address for your customers?

Core Problem:

Manual verification is time-consuming and highly inefficient.

Key issues:

- Increasing fraudulent claims
- Financial losses
- Slow checks
- Pattern identification difficulty
- Human error

Goal: System must identify fraud automatically.

9. PROBLEM ROOT CAUSE

RC

What is the real reason that this problem exists? What is the back story?

Causes:

- False accident claims
- Exaggerated damage
- Duplicate claims
- Property damage & fraud rings

Fraudsters continuously exploit strengths in manual verification systems.

Solution: ML analyzes patterns to identify suspicious activity.

7. BEHAVIOUR

BE

What does your customer do to address the problem and get the job done?

Current Actions:

- Investigating suspicious claims manually
- Checking history
- Reviewing accident reports
- Analyzing patterns
- Consulting investigators

Issue: These processes are far too slow and inefficient for large datasets.

FOCUS ON J&P TAP INTO BE, UNDERSTAND RC

IDENTIFY STRONG TR & EM

3. TRIGGERS

TR

What triggers customers to act?

- Rising number of fraudulent claims & reported losses
- Scam news reports
- Increased processing time
- Demand for automated processing & government regulations

MOTIVATION TO ADOPT AUTOMATED SYSTEMS.

4. EMOTIONS: BEFORE / AFTER

EM

How do customers feel when they face a problem and afterwards?

BEFORE:

Frustrated, concerned, lack of confidence, stressed.

AFTER:

More confident, faster processing, reduced losses, improved efficiency.

Result: Confidence and control.

10. YOUR SOLUTION

SL

Fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

Insurance Fraud Detection System using ML

Analyzes data, seamlessly predicts whether claims are fraudulent or legitimate.

Components:

- Data preprocessing & EDA
- Feature engineering
- Model training/evaluation
- Deployment (Flask)

Algorithms used:

- Decision Tree
- Random Forest
- KNN
- Logistic Regression
- Naïve Bayes
- SVM

BENEFITS:

- Faster detection & reduced losses
- Automated verification
- Improved accuracy

8. CHANNELS OF BEHAVIOUR

CH

What kind of actions do customers take online/offline?

8.1 ONLINE

- Insurance software
- Analytics dashboards
- Cloud platforms

8.2 OFFLINE

- Investigation teams
- Claim review meetings
- Departments

ML system integrates seamlessly with these existing channels.

IDENTIFY STRONG TR & EM